

1. Illustrate the operation of FIND-MAXIMUM-SUBARRAY on the array $A = \langle -7, 9, -5, -4, 5, 6, 3, -1 \rangle$.

2. Use the master method to give tight asymptotic bounds for the following recurrences.

(a) $T(n) = 81T(n/3) + n^2.$

(b) $T(n) = 81T(n/3) + n^4.$

(c) $T(n) = 81T(n/3) + n^6.$